

Muhammad Atiq

Data Engineer | Python & Backend Developer | Data Pipelines, Cloud & AI Systems

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PROFESSIONAL SUMMARY

Software engineer with 3+ years of professional experience building production-grade Python backends, REST APIs, and data pipelines, now specialising in data engineering through an MSc in Data Science. Hands-on with the modern data stack — Docker, AWS, FastAPI, and LLM's — with three live, deployed data systems demonstrating end-to-end pipeline design, orchestration, and analytics. AWS-certified, with a track record of shipping reliable, well-tested services in agile, cross-functional teams.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript / TypeScript, C++, R, Bash

Data Engineering & Pipelines: ETL / ELT, Data Modelling, Data Validation, Batch Scheduling (APScheduler)

Databases: PostgreSQL, SQLite, MongoDB, Redis

Backend & APIs: FastAPI, Flask, REST API Design, SQLAlchemy, Pydantic, Celery, Node.js / Express

Cloud & DevOps: AWS, Docker, Linux, Git / GitHub, CI/CD

Data Science & Analytics: pandas, NumPy, scikit-learn, Streamlit, Plotly, Machine Learning, Gemini API

Practices: Agile / Scrum, Asynchronous Systems, Automated Testing, Object-Oriented Programming

PROJECTS

Market Data Reconciliation Pipeline | Python · Apache Airflow · dbt · DuckDB · FastAPI · Streamlit · Docker

GitHub: github.com/Umrii/market-data-reconciliation-pipeline

- Built an end-to-end data engineering pipeline reconciling equities market data from multiple sources, orchestrated with Apache Airflow and modelled with dbt on a DuckDB warehouse.
- Implemented ingestion, transformation, and reconciliation logic to detect and surface cross-source discrepancies, improving data quality and trust.
- Exposed reconciled data through a FastAPI service and an interactive Streamlit dashboard for monitoring and analysis.
- Containerised the full stack with Docker for reproducible local and cloud deployment.

Carbon Market Tracker | Python · FastAPI · SQLite · APScheduler · Streamlit · Plotly · Gemini API

GitHub: github.com/Umrii/carbon-market-tracker · Live: carbon-market-tracker.onrender.com

- Built an end-to-end pipeline fetching live EU ETS carbon allowance (EUA) prices from the European Energy Exchange (EEX), with a SQLite storage layer using upsert logic and daily scheduling via APScheduler.
- Computed rolling analytics (7- and 30-day moving averages, 20-day annualised volatility) and automated $\pm 3\%$ price-movement alerts.
- Added a Gemini API layer generating natural-language commentary and market summaries on top of the data.
- Served data via a FastAPI REST API and an interactive Streamlit dashboard with Plotly candlestick charts; deployed live on Render.

Scout — AI Expert-Sourcing Agent | Python · FastAPI · Gemini API · Pydantic · Evaluation Harness

Live: scout-ibs2.onrender.com

- Built an AI agent that sources and extracts structured expert profiles from unstructured web data, exposed via a FastAPI service and deployed live on Render.
- Designed a versioned evaluation harness measuring extraction accuracy and hallucination rate against a live LLM, improving extraction accuracy from roughly 79% to 90% and halving the hallucination rate across iterations.
- Applied prompt engineering and structured-output validation (Pydantic) to produce reliable, machine-readable results.

Classification System Under Class Imbalance | R · RStudio · Random Forest · Logistic Regression

- Built an end-to-end ML system predicting online shopper purchase intent, covering preprocessing, feature engineering, model training, and evaluation.
- Applied class-imbalance techniques (resampling / class weighting) and evaluated with ROC-AUC, F1-score, and precision-recall curves rather than raw accuracy.

PROFESSIONAL EXPERIENCE

Python Developer — Hoboetech

Mar 2025 – Sep 2025

Lahore, Pakistan

- Built and maintained Python data ingestion and cleaning pipelines that delivered analysis-ready datasets for downstream analytics and reporting.
- Introduced data validation, monitoring, and fault-handling (retry logic and failure recovery) that reduced pipeline failures and improved data integrity in production.
- Increased data workflow efficiency through automation and optimisation, cutting manual processing effort.
- Partnered with cross-functional teams to diagnose performance bottlenecks and improve system maintainability and scalability.

Python Developer — Cyber Evangelists

Jul 2023 – Feb 2025

Lahore, Pakistan

- Designed and deployed 10+ production-grade REST APIs with FastAPI, serving real-time data to security-scanning and asset-analysis products handling 1,000+ daily requests.
- Built distributed, asynchronous processing systems with Celery and Redis for event-driven and long-running workloads, improving throughput and responsiveness.
- Optimised SQL queries and indexing to reduce API latency by 12%, improving data-retrieval efficiency for business operations.
- Containerised and deployed backend services with Docker across multiple environments, improving deployment consistency and reliability.
- Automated vulnerability-scanning workflows integrating tools such as OWASP ZAP, Nmap, and WHOIS, reducing manual security-analysis effort.
- Delivered backend features in an agile, cross-functional team while following secure-coding practices.

Software Engineer Intern — Bitsclan IT Solutions

Jun 2022 – Jun 2023

Lahore, Pakistan

- Developed 2 full-stack prototype web applications, turning product concepts into working demos for stakeholder review and validation.
- Built responsive React.js interfaces with reusable components, improving UI consistency and reducing development time.
- Designed backend services and REST APIs with Node.js / Express and MongoDB to support application logic and data flow.
- Accelerated early-stage prototyping cycles through reusable components, shortening product-validation timelines.
- Contributed to code reviews and debugging sessions, adopting modern software-development practices.

EDUCATION

MSc Data Science (with Advanced Practice) — Northumbria University

Sept 2025 – June 2027

Newcastle upon Tyne, UK · Relevant modules: Big Data & Cloud Computing, Advanced Databases, Machine Learning, Statistical Programming.

BSc Computer Science — FAST — National University of Computer & Emerging Sciences

2019 – 2023

Faisalabad, Pakistan

CERTIFICATIONS

- AWS Certified AI Practitioner (AIF-C01) — 2026
- AWS Certified Cloud Practitioner (CLF-C02) — In Progress (expected 2026)

LEADERSHIP & EXTRACURRICULAR

Head of Guest Protocol — FAST-NUCES, Pakistan

[2023]

- Led guest-protocol operations for a university olympiad hosting 5,000+ attendees, coordinating hospitality, logistics, and stakeholder management across the event.

Outgoing Global Exchange Team Lead — AIESEC in Pakistan

Feb 2022 – May 2023

- Coordinated international exchange programmes and led cross-cultural teams, developing stakeholder management and communication skills.

Teaching Assistant — FAST-NUCES, Pakistan

Aug 2021 – Jun 2022

- Supported delivery of lectures and tutorials, designed and graded assessments, and mentored students one-on-one, strengthening technical communication.

PART-TIME / ADDITIONAL EXPERIENCE

Team Member — Gainford Group (Vermont Hotel)

Dec 2025 – Present

Newcastle upon Tyne, UK

- Deliver customer-facing hospitality support in a fast-paced, time-sensitive environment while maintaining accuracy and professionalism.
- Balance part-time work alongside full-time postgraduate study, demonstrating strong time management and reliability.